

SOVEREIGN MASTER MANUAL

CLOSED-LOOP HUB OPERATIONS

A Comprehensive Operational Runbook for Decentralized Food and Energy
Infrastructure Management

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PRESTIGE COSMIC VENTURE

Core Competencies Overview

To guarantee autonomous operation, station operators must master six critical engineering domains:

Domain Ref	Competency Specialty	Target Milestone Target
MOD-01	Poultry Sprint Barn Management	Zero-grid brooding and batch velocity tracking
MOD-02	Swine Growth & Isolation Pods	Biosecurity containment and hydraulic gutter flushes
MOD-03	Anaerobic Biogas & Generation	Methane synthesis, gas scrubbing, and co-generation
MOD-04	Off-Grid Inverter & Storage	30kW Hybrid routing and LFP battery life optimization

The Closed-Loop Ecosystem

Interconnected Utility Logic

- ✓ **Waste Harvesting:** Manure slurry is captured straight from the barns to eliminate ammonia off-gassing.
- ✓ **Bacterial Digestion:** Organic slurry fuels anaerobic systems, creating structural methane gas.
- ✓ **Thermal Protection:** Methane combustion outputs electrical voltage and recycled floor steam loops.



Sovereign Goal

The farm processes inputs into energy, shielding livestock from Highveld winter freezes without reliance on outside utilities.

Poultry Intake & Brooding Setup

Day 1 Critical Checklists

- ✓ **Pre-Heat Target:** Warm concrete floors to 32°C utilizing co-generation heat 24 hours prior to delivery.
- ✓ **Curtain Setting:** Set manual winch PVC curtains over dwarf walls to lock out ground-level freezing drafts.
- ✓ **Hydration Priming:** Provide warm, sanitized water mixed with vital electrolytes to stabilize arrivals.



Novice Rule

Cold floors stunt chick development. Continuous thermal checks during the initial 48 hours are mandatory.

Poultry Batch Feed Transitions

Feed allocations must match structural growth phases to protect high-velocity turnover curves:

Batch Phase	Day Horizon	Milling Blend Focus	Target Physical Behavior
Starter	Days 01 – 14	22% Crude Protein fine crumbles	Accelerated skeletal bone structure growth
Grower	Days 15 – 28	19% Crude Protein mid-size pellets	Muscle mass and lean tissue expansion
Finisher	Days 29 – 42	18% Crude Protein high-energy pellets	Optimal target market weight attainment

Poultry Harvest & Barn Evacuation

Depopulation & Sanitization

- ✓ **Feed Isolation:** Withdraw feed 8 hours before bird catching to ensure pristine transport processing.
- ✓ **Catching Conditions:** Adjust side PVC canvas down to work in low-light environments, keeping birds calm.
- ✓ **Manure Scraping:** Remove all litter immediately to stop internal ammonia gas buildups.



Turnaround Guard

Allow a rigid 14-day dry air-gap window between batches for complete chemical sanitization and pathogen breakdown.

Swine Onboarding & Stocking

Finisher Barn Entry Protocol

- ✓ **Weight Sorting:** Group arriving weaners into pens by size to prevent dominant animal injury.
- ✓ **Drinker Calibration:** Set automatic nipple drinker line heights to align with the pigs' shoulder levels.
- ✓ **Slat Inspection:** Confirm polypropylene floor panels are locked tight over the sub-floor channels.



Density Limits

Maintain an exact 1.2 m² allocation per hog across the 16 pens to prevent territorial stress and maintain stable growth paths.

The 4-Pod Nursery Isolation System

Isolation Element	Standard Field Protocol	Biosecurity Security Impact
Airtight Barriers	Keep individual entry doors closed tightly at all times.	Halts the drift of airborne pathogens between chambers.
Decoupled Foot Baths	Step into dedicated viral chemical washes before passing doors.	Eliminates cross-contamination tracking via boots.
Isolated Drainage	Keep pod drain valves shut during targeted observation.	Prevents moisture leaks from backing up adjacent pipelines.

Gutter Flush & Slurry Redirection

Hydraulic Waste Operations

- ✓ **Interval Routine:** Execute a full concrete channel flush every 48 hours to maintain pen hygiene.
- ✓ **Sluice Gate Operation:** Lift concrete release gates rapidly to utilize maximum hydraulic flushing heads.
- ✓ **Pre-Scrape Check:** Verify internal floor grids are clear before releasing the waste channels.



Gravity Transit

The channels rely entirely on built-in slopes to carry waste slurry into the continuous digestion systems without mechanical pumps.

Biogas Intake & Slurry Balancing

Digester Core Optimization

- ✓ **Dilution Ratios:** Mix raw swine manure with borehole water at a strict 1:1 fluid weight ratio.
- ✓ **Solids Threshold:** Maintain overall total volatile solids down at an 8% to 10% operational consistency.
- ✓ **Feeding Cadence:** Charge the reactor tank at set times daily to prevent shocking the biological bacteria.



Acid Alert

Overfeeding causes internal souring. Monitor effluent pH levels constantly, keeping them within a stable 6.8 to 7.4 range.

Methane Scrubbing & H2S Filtration

Filter Component	Operator Action Step	Corrosion Defense Goal
Iron Swarf Traps	Inspect and shake loose reactive binding iron media every 30 days.	Binds gaseous Hydrogen Sulfide (H_2S) to protect down-line metal pipes.
Condensate Drains	Purge water knockout valves manually every morning.	Removes water vapor to prevent pipeline water-lock.
Pressure Bubblers	Maintain fluid levels up to the marked indicator lines.	Acts as a safety backup to prevent reverse gas back-flashes.

Co-Generation Engine Management

Power & Heat Extraction

- ✓ **Manifold Injection:** Route scrubbed methane gas into the modified generator line at 15-22 kPa.
- ✓ **Exhaust Interception:** Divert hot engine exhaust gas into the stainless steel heat exchanger matrix.
- ✓ **Steam Distribution:** Route recovered hot steam through insulated pipe networks into nursery floors.

Run Limits

Synchronize generator runtimes with peak evening heating demands to maximize overall thermal efficiency across the farm.

30kW Solar & Inverter Interface

Daily Inverter Operations

- ✓ **MPPT Curve Check:** Confirm both high-voltage DC string lines show matching active tracking inputs.
- ✓ **Isolation Relays:** Verify automated open-loop switchgear status flags show green for island mode.
- ✓ **Dust Clearance:** Clean solar panel surfaces weekly during dry highveld winters to protect generation.



Safety Rule

Never open high-voltage DC fuse boxes while the system is generating under full load. Disconnect the primary AC breakers first.

30kWh Lithium BESS Stewardship

BESS Diagnostic Data	Target Operational Zone	Stewardship Action Step
State of Charge (SoC)	20% up to 95% maximum bounds	Limit overnight depth of discharge to maintain battery health.
Cell Balance Range	Less than 15mV deviation variance	Engage monthly trickle cycles to let BMS equalize cells.
Enclosure Temperature	15°C up to 30°C optimal range	Keep cabinet fan filters clear to prevent thermal stress.

LoRaWAN Sensor Network Care

Edge Sensor Maintenance

- ✓ **Gateway Health:** Check the central IP67 gateway status dashboard for active data loops daily.
- ✓ **Ammonia Calibration:** Clean internal NH_3 sensor filters every 60 days to prevent fouling.
- ✓ **Battery Monitoring:** Swap internal node batteries when telemetry drops past the 2.2V threshold.



Telemetry Anchors

Sensors are the system's eyes. Unread alerts expose livestock batches to rapid, unmonitored climate drops.

Hydrological Borehole Operations

Water Transfer Framework

- ✓ **DC Pump Control:** Configure solar controllers to activate the submersible pump during solar peaks.
- ✓ **Tower Management:** Check gravity JoJo tank mechanical float valves for smooth shutoffs daily.
- ✓ **Reserve Buffer:** Keep overall tank capacities above the 5-day emergency storage line.



Pressure Delivery

Elevated storage towers maintain line pressure via gravity, keeping drinker nipples open even during outages.

Inline Purification & Sterilization

Purification Stage	Operational Protocol Checklist	Pathogen Defense Impact
Kinetic Disc Bank	Open manual drain valves weekly to flush trapped sand sediment.	Keeps down-line automation valves clear of blockages.
Inline UV Chamber	Verify glass sleeves are clean and look for the blue glow daily.	Neutralizes waterborne viral chains on contact.
Proportional Doser	Mix chlorine baseline stocks to provide 2 ppm residual levels.	Prevents bacteria and biofilms from building inside piping.

Sovereign Financial Accounting

Micro-Accounting Log Metrics

- ✓ **Tariff Baseline:** Value internal power usage at a fixed ****1.80 ZAR/kWh**** in all models.
- ✓ **Batch Ledger Sync:** Log feed consumption against animal weights weekly to calculate feed conversions.
- ✓ **Yield Projections:** Tag all sales under regional, off-grid premium distributions.



Economic Ledger

The 1.80 ZAR/kWh rule quantifies energy cost savings, demonstrating the microgrid's true returns to tracking auditors.

Emergency Response Protocols

Incident Context	Immediate Core Action Step	Secondary Safety Action
Total Microgrid Failure	Isolate the LFP battery bank and switch main inverter breakers off.	Start the biogas generator manually to run emergency lines.
Drastic Water Drop	Close primary barn header drops to protect remaining tower reserves.	Inspect pipe runs to locate and isolate the structural leak.
Extreme Ammonia Spikes	Hoist side PVC weather curtains manually to clear air paths.	Initiate emergency floor scrapes and inspect the slurry gates.

TRAINING MANUAL COMPLETE

SOVEREIGNTY THROUGH KNOWLEDGE MASTERY

Lwandile Engineering Projects. Secures this Manual inside our on-site terminal. Operators to ensure review and log signatures - weekly

Secunda & Kinross Training Framework | Just Energy Transition Implementation Portfolio

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